

## DET-014 | PsExec Remote Execution Detection

ID: DET-014

MITRE ID:T1569.002

Data Source: DeviceProcessEvents

Severity: High

Tactic:Lateral Movement

Category:LATERAL MOVEMENT

### ADX MAPPING LOGIC:

Mapping: Source = 'tool/binary name (psexec)', State = 'remote target hostname'. We use parse operator to extract structured fields from an unstructured string — the exact technique used for parsing command-line arguments in real Sentinel.

OBJECTIVE: Practice the parse operator — used to extract structured data from unstructured command-line strings. Real Sentinel: extract remote hostname from PsExec command-line pattern like 'psexec \\hostname -u admin cmd.exe'.

### KQL QUERY (RUNNABLE ON ADX FREE TIER)

StormEvents

```
| where StartTime > ago(365d)
| extend RawCommandLine = strcat(
EventTypes, " target:", State, " source:", Source,
" damage:", tostring(DamageProperty))
| parse RawCommandLine with
ProcessName:string " target:" RemoteTarget:string
" source:" AttackerSource:string
" damage:" DamageStr:string
| extend
IsRemoteExec = EventType in ("Tornado","Waterspout"), // = psexec/paexec
DamageVal = toint(DamageStr),
RiskScore = iff(EventType == "Tornado", "CRITICAL", "HIGH")
| where IsRemoteExec == true
| project StartTime,
ProcessName,RemoteTarget,AttackerSource,
DamageVal,RiskScore,
RawCommandLine
| order by DamageVal desc
| take 20
```

### INVESTIGATION STEPS

1. Identify source and destination of the PsExec execution.
2. Check if this is authorised sysadmin activity (change ticket).
3. Review what command was executed on the remote host.
4. Check if PSEXESVC.exe was installed as a service on the target.

## RESPONSE ACTIONS

- If unauthorised: isolate both source and destination hosts.
- Remove PSEXESVC service from target hosts.
- Review all commands executed via PsExec session.
- Hunt for PsExec activity across the entire environment.

## Output from Above KQL Query:

```
1 StormEvents
2 | where StartTime > ago(365d)
3 | extend RawCommandLine = strcat(EventType, " target:", State, " source:", Source,
4 | damage:", toString(DamageProperty))
5 | parse RawCommandLine with
6 ProcessName:string " target:" RemoteTarget:string
7 " source:" AttackerSource:string
8 " damage:" DamageStr:string
9 | extend
10 IsRemoteExec = EventType in ("Tornado","Waterspout"), // = psexec/paexec
11 DamageVal = toint(DamageStr),
12 RiskScore = iff(EventType == "Tornado", "CRITICAL", "HIGH")
13 | where IsRemoteExec == true
14 | project StartTime, ProcessName, RemoteTarget, AttackerSource,
15 DamageVal, RiskScore, RawCommandLine
16 | order by DamageVal desc | take 20
```

Table 1

Add visual

Stats

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| StartTime       | ProcessName | RemoteTarget | AttackerSource | DamageVal | RiskScore | RawCommandLine |
|-----------------|-------------|--------------|----------------|-----------|-----------|----------------|
| No Rows To Show |             |              |                |           |           |                |

## KQL Detection Query in Sentinel:

DeviceProcessEvents

```
| where TimeGenerated > ago(1h)
| where FileName in~ ("psexec.exe", "psexec64.exe", "paexec.exe")
or ProcessCommandLine has _any (
"psexesvc", "\\admin$", "\\ipc$"
)
or (FileName =~ "PSEXESVC.exe")
| project
TimeGenerated, DeviceName, AccountName,
FileName, ProcessCommandLine,
InitiatingProcessFileName,
RemoteDevice = extract(
@"\\([^\s]+)", 1, ProcessCommandLine
)
| order by TimeGenerated desc
```